Chapter 9 — Acid-Base Chemistry

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. pH equals: (a) $[H^+][OH^-]$ (b) $\log[H^+]$ (c) $-\log[OH^-]$ (d) $-\log[H^+]$

2. pOH equals: (a) $\log[OH^-]$ (b) $-\log[H^+]$ (c) $[H^+][OH^-]$ (d) $-\log[OH^-]$

3. pH + pOH at 298 K: (a) 1 (b) 14 (c) 7 (d) 0

4. 0.01 M HCl pH: (a) 7 (b) 1 (c) 12 (d) 2

5. Ka measures: (a) Solubility (b) Buffer mass (c) Base strength (d) Weak acid strength

6. Acidic buffer example: (a) HCl/NaCl (b) NaOH/NaCl (c) CH_3COOH/CH_3COONa (d) H_2O/NaCl

7. Common ion effect suppresses: (a) Precipitation only (b) Redox (c) Ionization (d) Combustion

8. Ksp predicts: (a) Precipitation (b) Colour (c) Density (d) pH only

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. How does dilution affect the pH of a strong acid?

2. Why does a weak acid have a higher pH than a strong acid of the same concentration?

3. Why does a buffer resist pH change?

4. How does adding sodium acetate affect acetic acid ionization?

5. How does Ka indicate acid strength?

6. How can Ksp predict whether a precipitate forms?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Explain pH, pOH and calculations for strong acids and bases. **(4)**

(b) Explain Ka and calculations for weak acids. **(4)**

Q.4 (a) Explain common ion effect and buffer solutions. **(4)**

(b) Explain Ksp, solubility and prediction of precipitation. **(4)**

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